Retail Flow as a Hidden Risk Factor

real__full__standard__fast__w60__high

Parameter	Value
Data Source	Real data — FINRA off-exchange proxy + SPY/VIX (yfinance)
Period	Full sample (2018–2024)
Horizons	Standard horizons (5, 10, 20d)
Walk-Fwd	Fast refit (250d initial train, 10d step)
Reflex Win	60d reflex window (baseline)
Ridge Alpha	Heavy regularization (alpha=5.0)

Walk-Forward OOS Performance

Horizon	Spec	MAE	RMSE	OOS R² vs naive
5	ablation_reflex_gap_stress	0.0558	0.0931	0.2028
5	ablation_retail_share_z	0.0572	0.0933	0.2001
5	ablation_short_volume_ratio	0.0565	0.0937	0.1929
5	har_rv	0.0589	0.1007	0.0683
5	har_rv_vix	0.0558	0.0931	0.2039
5	har_rv_vix_retail	0.0573	0.0935	0.1963
5	naive_persistence	0.0658	0.1043	0.0000
10	ablation_reflex_gap_stress	0.0541	0.0915	0.2247
10	ablation_retail_share_z	0.0561	0.0920	0.2157
10	ablation_short_volume_ratio	0.0552	0.0925	0.2074
10	har_rv	0.0578	0.0996	0.0804
10	har_rv_vix	0.0540	0.0911	0.2299
10	har_rv_vix_retail	0.0563	0.0925	0.2066
10	naive_persistence	0.0611	0.1039	0.0000
20	ablation_reflex_gap_stress	0.0540	0.0949	0.2978
20	ablation_retail_share_z	0.0574	0.0973	0.2621
20	ablation_short_volume_ratio	0.0552	0.0959	0.2841
20	har_rv	0.0574	0.1017	0.1947
20	har_rv_vix	0.0539	0.0947	0.3017
20	har_rv_vix_retail	0.0579	0.0980	0.2524
20	naive_persistence	0.0614	0.1133	0.0000

Incremental Value: retail features vs. har_rv_vix

parison	Horizon	N Obs	Hac Lag	Mean Sq Err Diff	Oos R2 Vs Bench	Dm Stat	Dm Stat Hln	P Value	P Naive Ttest	Challenger B
v_vix_retail vs har_rv_vix	5	1301	4	0.0001	-0.0095	0.5035	0.5018	0.6159	0.4206	False
on_retail_share_z vs har_rv_vix	5	1301	4	0.0000	-0.0048	0.2816	0.2806	0.7790	0.6560	False
on_reflex_gap_stress vs har_rv_vix	5	1301	4	0.0000	-0.0014	0.4580	0.4564	0.6482	0.6826	False
on_short_volume_ratio vs har_rv_vix	5	1301	4	0.0001	-0.0138	1.2824	1.2779	0.2015	0.0480	False

v_vix_retail vs har_rv_vix	10	1296	9	0.0003	-0.0303	1.2355	1.2265	0.2202	0.0242	False
on_retail_share_z vs har_rv_vix	10	1296	9	0.0002	-0.0185	0.8004	0.7946	0.4270	0.1572	False
on_reflex_gap_stress vs har_rv_vix	10	1296	9	0.0001	-0.0068	1.3640	1.3540	0.1760	0.0850	False
on_short_volume_ratio vs har_rv_vix	10	1296	9	0.0002	-0.0293	2.5980	2.5789	0.0100	0.0000	False
v_vix_retail vs har_rv_vix	20	1286	19	0.0006	-0.0706	1.7119	1.6859	0.0920	0.0000	False
on_retail_share_z vs har_rv_vix	20	1286	19	0.0005	-0.0567	1.4991	1.4764	0.1401	0.0001	False
on_reflex_gap_stress vs har_rv_vix	20	1286	19	0.0000	-0.0055	1.4009	1.3797	0.1679	0.0044	False
on_short_volume_ratio vs har_rv_vix	20	1286	19	0.0002	-0.0252	2.1074	2.0755	0.0381	0.0000	False